ENVIRONMENTAL IMPACT ASSESSMENT

Proposed Second Padma Barrage

A comprehensive environmental impact assessment for policymakers, senior engineers, and development planners — examining ecological, hydrological, and socio-economic implications of the proposed second barrage on the Padma River, Bangladesh.

Hydrology

River flow & sediment dynamics

Ecology

Biodiversity & wetland impact

Communities

Displacement & livelihoods

Policy

Regulatory & governance framework

NATIONAL STRATEGIC INFRASTRUCTURE

Padma River Barrage Project

A government-funded initiative to construct a multi-purpose barrage across the Padma River near Hardinge Bridge, regulating monsoon flow to raise dry-season water levels and secure water for **70 million people across 19 districts**.

STRATEGIC OBJECTIVES

Five Pillars of National Impact

The barrage addresses Bangladesh's most critical water challenges through five interconnected objectives.

Water Security

Reliable dry-season supply for agriculture, households, and industry.

Ecological Restoration

Revive distributaries, counter salinity intrusion, and restore Sundarbans biodiversity.

Agricultural Growth

Expand irrigation, stabilize yields, and reduce rainfall dependency.

Economic Development

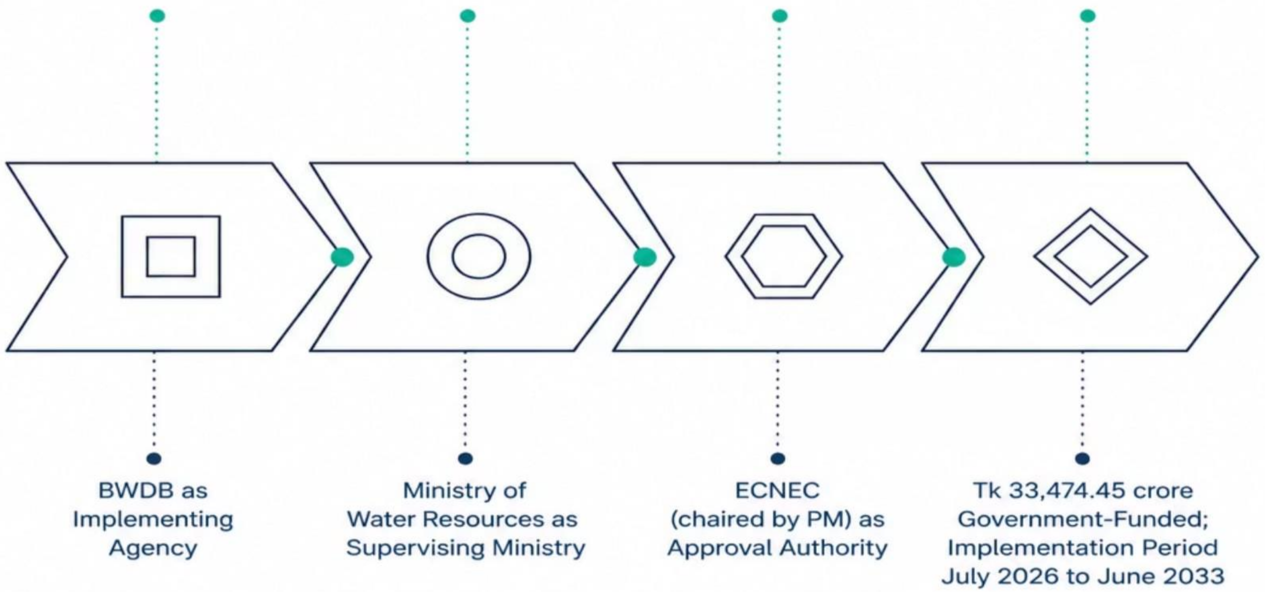
Enhance fisheries, navigation, and regional trade opportunities.

Transboundary Diplomacy

Strengthen Bangladesh's position in water-sharing negotiations with India and basin countries.

PROJECT FRAMEWORK

Implementation Structure & Rationale



Why Now

- **Farakka Impact:** Upstream diversions reduce dry-season flow to **15%** of Bangladesh's rightful share, driving severe salinity intrusion.
- **Sundarbans Crisis:** Rising salinity threatens mangrove ecosystems and livelihoods .The barrage is an "elixir of life" for the delta.
- **Climate Variability:** Increasing rainfall unpredictability makes controlled storage essential.
- **Technical Feasibility:** Studies since the 1960s confirm stable riverbanks and viable engineering solutions.

Location & Geographic Scale

- **SITE**
  Pangsha Upazila, Rajbari District, on the Padma River (Bangladesh's stretch of the Ganges)
- **POSITION RELATIVE TO FARAKKA**
  About 180 km downstream of India's Farakka Barrage in West Bengal
- **RIVER SYSTEMS RESTORED**
  Ichamati-Mathabhanga, Gorai-Madhumati, Chandana-Barasia, and Boral

SCALE OF IMPACT ACROSS BANGLADESH

37%
of Bangladesh's total geographical area

26
districts, 163 upazilas, 4 divisions

~70M
people (about one-third of the population) affected

Bangladesh's southwestern and northwestern Padma-dependent areas have faced decades of dry-season water scarcity linked to reduced upstream flows.

Source: Bangladesh Water Development Board / ECNEC approval, May 2026

Size of the Project

2.1 km
Main barrage length

78 spillway gates + 18 undersluices

2.9 bn m^3
Reservoir storage capacity

628 km^2 reservoir surface area

113 MW
Hydropower generation capacity

Power station at the barrage site

2.88 M ha
Farmland brought under irrigation

Net cultivable land served

Tk 50,444 cr
Total estimated cost

Phase 1 approved at Tk 34,608 cr

2026–2033
Construction to completion

Two-phase build-out

Source: Bangladesh Water Development Board; ECNEC approval, May 2026

Technical & Engineering Design

The proposed Padma Barrage Project is designed as a multi-purpose river regulation structure that will store excess monsoon water and release it in a controlled manner during dry season.

Overall Structure

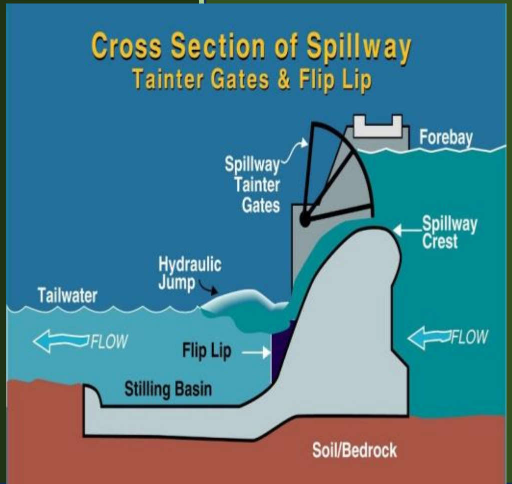
The barrage will be constructed across the Padma River at Charjiguri in Pangsha Upazila, Rajbari District. The main structure is expected to have a length of approximately 2.1 km and will contain 78 spillway gates to regulate river flow.

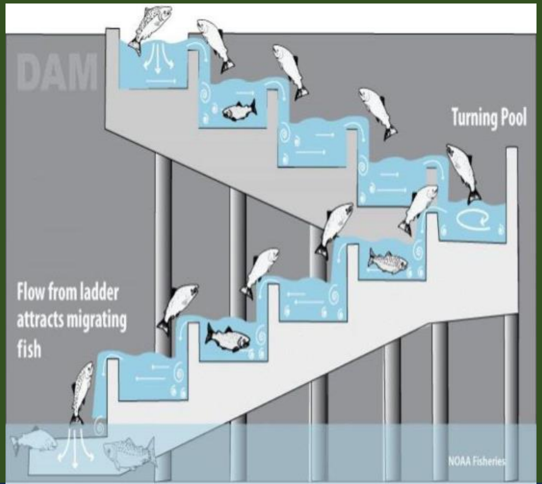
Engineering parameter	Proposed specification
Barrage length	2.1 km
Spillway gates	78
Reservoir capacity	~2.9 billion m^3
Reservoir surface area	~628 km^2
Hydropower capacity	113 MW
Primary functions	Irrigation, Navigation, Power

STRUCTURE COMPONENTS

The proposed Padma Barrage will consist of several integrated engineering structures designed to regulate river flow and support multiple functions:

- **Spillway gates (78):** Control the flow of water and maintain the required upstream water level.
- **Undersluices:** Allow excess water and sediment to pass through, helping reduce siltation behind the barrage.
- **Navigation locks:** Enable boats and cargo vessels to move across the barrage without interrupting river transport.
- **Fish passages:** Provide migration routes for fish to reduce the barrier effect on aquatic biodiversity.
- **Embankments and guide bunds:** Protect surrounding land from erosion and stabilize the river channel near the barrage.





HYDROPOWER COMPONENTS

The barrage includes a 113 MW hydropower component integrated with the water-release system.

- Water released through the barrage will pass through turbines.
- The turbines will convert the kinetic energy of flowing water into electricity.
- Power generation is considered a co-benefit, while irrigation and water regulation remain the primary objectives.

DESIGN PURPOSE CODES

I Irrigation
N Navigation
P Power

Seasonal Operation Of the Barrage

“ During the monsoon, the barrage stores and regulates excess river water. During the dry season, the stored water is released gradually to maintain downstream water levels."

This dual-season operating principle is the core function of the barrage. By capturing peak monsoon flows, the structure prevents downstream flooding while building a reserve that can be drawn upon during the dry months to sustain irrigation, navigation, ecological health, and hydropower generation throughout the year.

Monsoon Season — Store Excess Water

When monsoon rainfall swells the river beyond normal capacity, the barrage gates are managed to capture and hold the surplus flow. This prevents uncontrolled flooding downstream while accumulating a critical water reserve for later use.

Dry Season — Controlled Release

As rainfall diminishes and river levels drop, the barrage releases stored water in a controlled and graduated manner. Downstream water levels are maintained at levels sufficient for irrigation, navigation, ecological flow, and hydropower generation.

Year-Round Water Security

Together, these two operating modes transform an unpredictable seasonal river into a regulated water resource — delivering reliable flow to communities, agriculture, and ecosystems that depend on it throughout the entire year.

Cost, Financing & Institutional Framework

Proposed Padma Barrage Project

Project Investment

- Phase I: Tk 34,497 crore (~US$2.8B)
- Total estimate: Tk 50,443 crore
- Major national water infrastructure project

Financing Strategy

- Primarily domestic financing
- ADP & Government Budget
- Treasury resources
- Domestic borrowing (if needed)

Institutional Framework

- Executing Agency: BWDB
- Supervising Ministry: Ministry of Water Resources
- ECNEC approval (2026)
- BWDB to operate & maintain

KEY STAKEHOLDERS

BWDB (PROJECT OWNER & OPERATOR)

- **Ministry of water resources**
- **Ministry of Planning & ECNEC**
- **Local govt. & district administration**
- **Affected Communities** (Farmers, Fishermen, etc)
- **Environmental experts & NGOs**
- **International observers** (e,g.International Farakka Committee)

Why This Investment Matters

- Improve dry-season irrigation
- Reduce salinity intrusion
- Enhance river navigation & hydropower
- Support sustainable water management

What is Environmental Impact Assessment (EIA)?

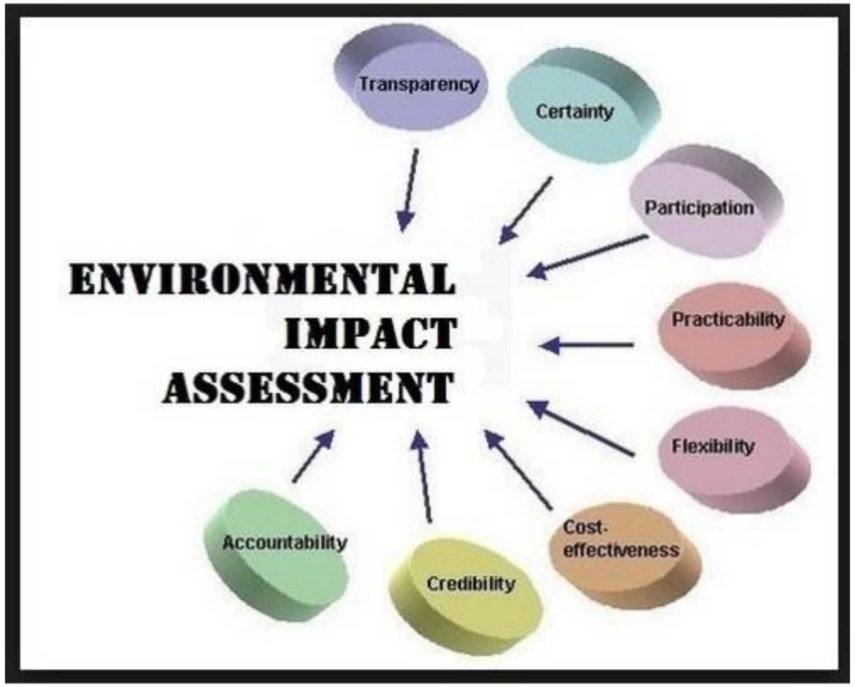
Definition: A systematic process to identify, predict, and evaluate the environmental and social impacts of a proposed project before approval.

Sustainable Development: Ensures development meets present needs without compromising future generations.

Informed Decision-Making: Provides decision-makers with scientific evidence to weigh trade-offs.

Public Participation: Legally mandates community and stakeholder involvement in the planning process.

Legal Compliance: Required under Bangladesh Environment Conservation Act and international best practices.



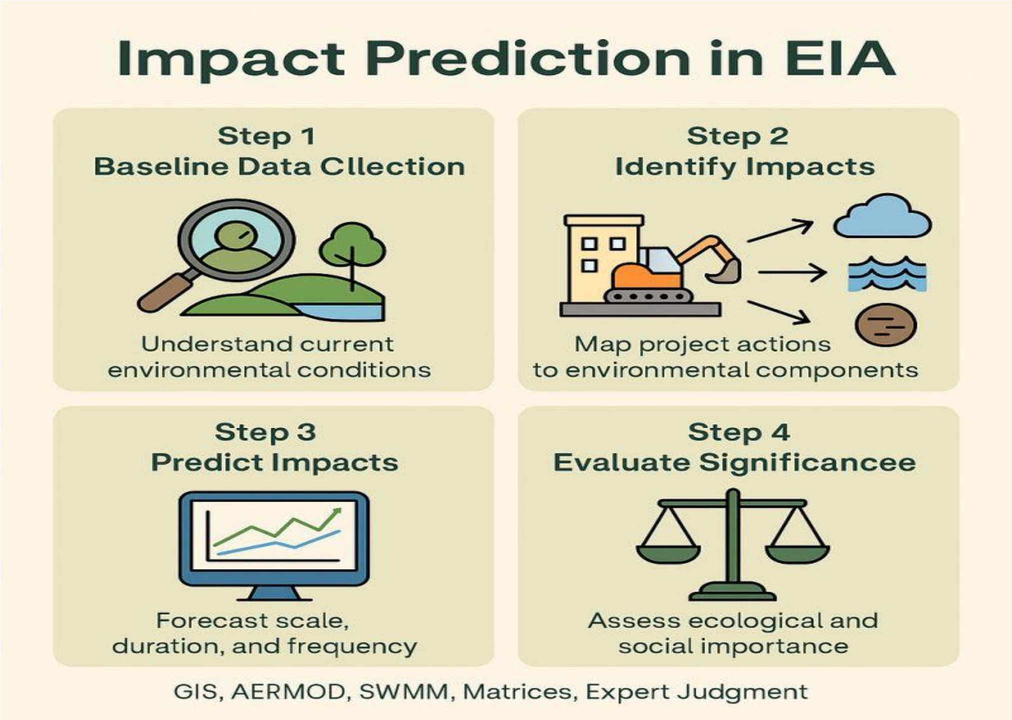
EIA Methodology: Data Collection & Impact Prediction

Baseline Data Collection: Surveys of hydrology, water quality, ecology, land use, and socioeconomic conditions in the project area.

Identification of IECs: Determining Important Environmental Components (IECs) most sensitive to project impacts.

Impact Prediction Modelling: Hydrological and sediment transport models using computerized simulation tools.

Expert Recommendation: Specialists have specifically called for advanced computerized modelling of sediment and hydrological impacts before implementation of the Padma Barrage.



EIA Methodology: Consultation, Mitigation & Management

Public Consultation: Structured stakeholder engagement with affected communities, local government, and civil society.

Mitigation Planning: Identifying measures to avoid, reduce, or compensate for identified negative impacts.

Environmental Management Plan (EMP):
A binding action plan specifying mitigation measures, responsibilities, timelines, and monitoring indicators.

Monitoring & Adaptive Management:
Continuous tracking of environmental parameters during construction and operation phases.

Positive Impacts: Water Security & Irrigation

Restoring dry-season flow across the Ganges distributary system

2.9 billion m³

of monsoon water retained and released back during the dry season

Distributaries to be revived

Gorai – Madhumati

Major southwest distributary; dredging of 135.6 km planned

Hisna – Mathabhanga

Re-excavation of 246.5 km of channel system

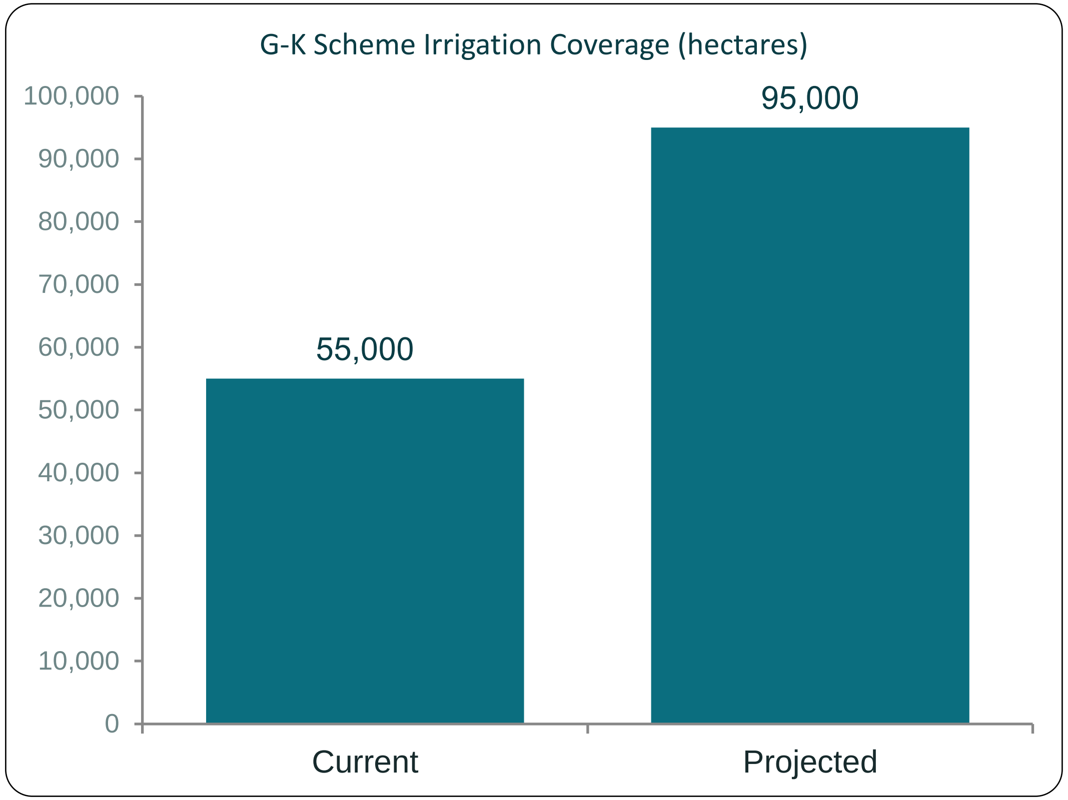
Chandana – Barashia & Ichamati

Offtake structures restore year-round flow

Sources: The Business Standard, Jan 2026; UNB, May 2026; BSS Special Stories, Jun 2026.

Expanding Surface Water Irrigation

The barrage strengthens the Ganges–Kobadak (G-K) scheme, Bangladesh's largest surface irrigation system



+ ~40,000 hectares of new irrigable farmland under the G-K scheme alone

Less reliance on iron-heavy groundwater

Local groundwater in the southwest is iron-rich and poorly suited for cultivation. Reliable surface-water supply from the barrage offers a cleaner, more consistent irrigation source.

Reduced dependence on diesel pumps

Farmers currently rely on costly, hard-to-maintain diesel-powered pumps to lift water. Gravity-fed surface supply lowers fuel costs and emissions for smallholders.

Water level maintained near 10 m to reliably feed the G-K canal network year-round

Sources: Dhaka Tribune, Jun 2026 (BWDB officials); BSS Special Stories, Jun 2026; The Daily Star, May 2026.

Boosting Crop Yields & Rural Livelihoods

Reliable dry-season water is expected to translate directly into farm output and household income

Higher cropping intensity

Assured surface water supports multiple harvests per year and expansion of dry-season boro rice cultivation.

Lower input costs

Gravity-fed canal water cuts farmers' fuel and pump-maintenance expenses compared with diesel lift-irrigation.

Stronger rural livelihoods

An estimated 70 million people across 19 districts stand to benefit from improved water security.

Wider regional reach

Supports the G-K Project and the proposed North Rajshahi Irrigation Project across the southwest and northwest.

Sources: BSS Special Stories, Jun 2026; Dhaka Tribune, Jun 2026; Prothom Alo, May 2026; UNB, May 2026.

Positive Impacts: Ecology, Navigation & Power

Reviving Ecology & River Navigation

Dry-season flow released from the barrage is designed to protect the Sundarbans and restore navigable rivers

Padma Barrage
regulated release

→

Gorai • Madhumati
Hisna • Chandana
distributary rivers

→

Sundarbans &
SW Coastal Belt
salinity reduction

Sundarbans & Salinity Control

- Targets salinity intrusion in Satkhira, Khulna, Bagerhat — freshwater supply for the Sundarbans ecosystem
- Addresses “top dying” of Sundarbans mangroves linked to excess salinity
- Two fish passes intended to keep migration routes open

River Navigability

- Restores flow/depth in Hisna-Mathabhanga, Gorai-Madhumati, Chandana-Barashia, Baral, Ichamati systems
- 135.6 km (Gorai-Madhumati) dredged + 246.46 km (Hisna) re-excavated
- Navigation locks built into the main barrage structure

Sources: The Financial Express (2026); The Business Standard (2026); The Daily Star (2026), “Tk 33,474cr Padma barrage set for govt nod”

Hydropower & Climate-Resilience Co-Benefits

Beyond water security, the barrage is framed as a clean-energy and climate-adaptation asset

113 MW

combined installed hydropower capacity

76.4 MW — main barrage plant

36.6 MW — Gorai off-take plant

A Broader Climate-Resilience Narrative

- Positioned by government as an adaptation response to sea-level rise, prolonged dry seasons and erratic rainfall across the Ganges delta
- Hydropower adds a renewable-energy co-benefit alongside irrigation and drought mitigation
- Feasibility estimates ≈ 0.45% contribution to GDP growth, framed as part of long-term water & energy security
- Experts note this is one piece of resilience — sediment and ecological trade-offs still need independent scrutiny (see Block 9–12)

Sources: Jagonews24 (2026), "ECNEC nods Padma Barrage project"; The Daily Star (2026); Drishti IAS (2026); The Climate Watch (2026)

Sediment Trapping Mechanism

≋ Velocity Drop & Deposition

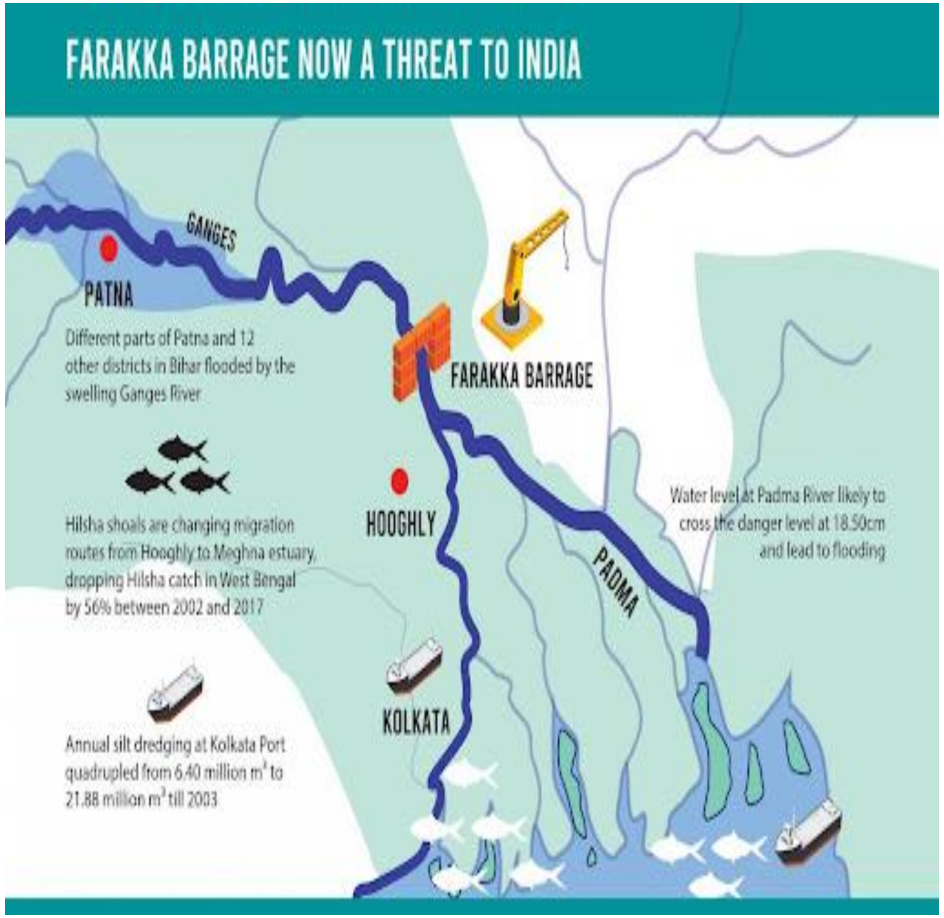
Constructing a barrage slows water flow upstream, turning a high-energy river into a reservoir. Reduced velocity forces suspended bedload and coarse silt to sink and get trapped behind the gates.

⚠ Farakka Barrage Precedent

Historical evidence from the Farakka Barrage shows that it already retains an estimated one-third to nearly half of the Ganges' total sediment load before reaching Bangladesh.

▼ Silt-Starved Release

Water released downstream loses its natural silt content, discharging "hungry water" that lacks essential nutrients required for delta maintenance.



Downstream Morphological Risks

400–600 Million Tonnes at Risk

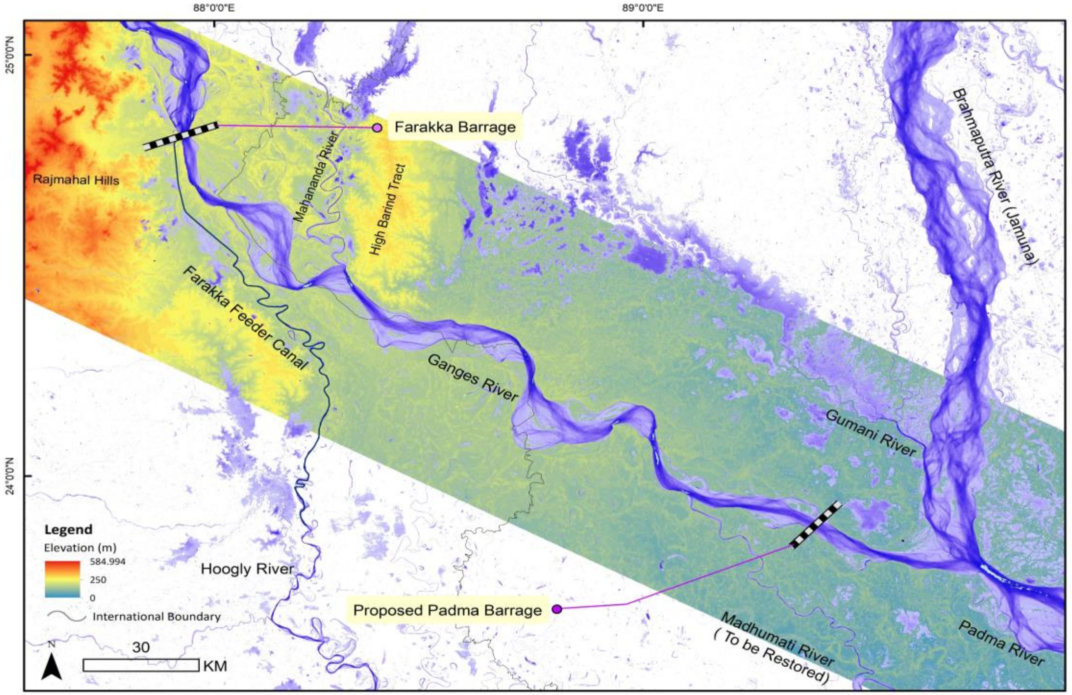
An additional barrage structure threatens to further reduce the crucial 400–600 million tonnes of sediment that currently reaching Bangladesh, threatening the long-term delta accretion process.

Reduced Floodplain Deposition

Loss of nutrient-rich sediment deposition during annual floods depletes natural soil fertility across downstream floodplains, increasing dependence on artificial fertilizers.

Channel Instability & Scouring

Clear, energy-dense water leaving the barrage gates scours the downstream riverbed, causing severe bank erosion and unpredictable channel course alterations.



Reservoir Silt & Heavy Metal Risk

Rapid Reservoir Siltation

Continuous trapping of incoming silt rapidly fills the upstream reservoir basin, progressively diminishing active storage volume and shortening overall barrage lifespan.

Heavy-Metal Trapping

Upstream industrial effluents and agricultural runoff bind to fine clay particles, concentrating toxic heavy metals (cadmium, lead, chromium) in trapped benthic sludge.

Benthic Toxicity & Food Chain Bioaccumulation

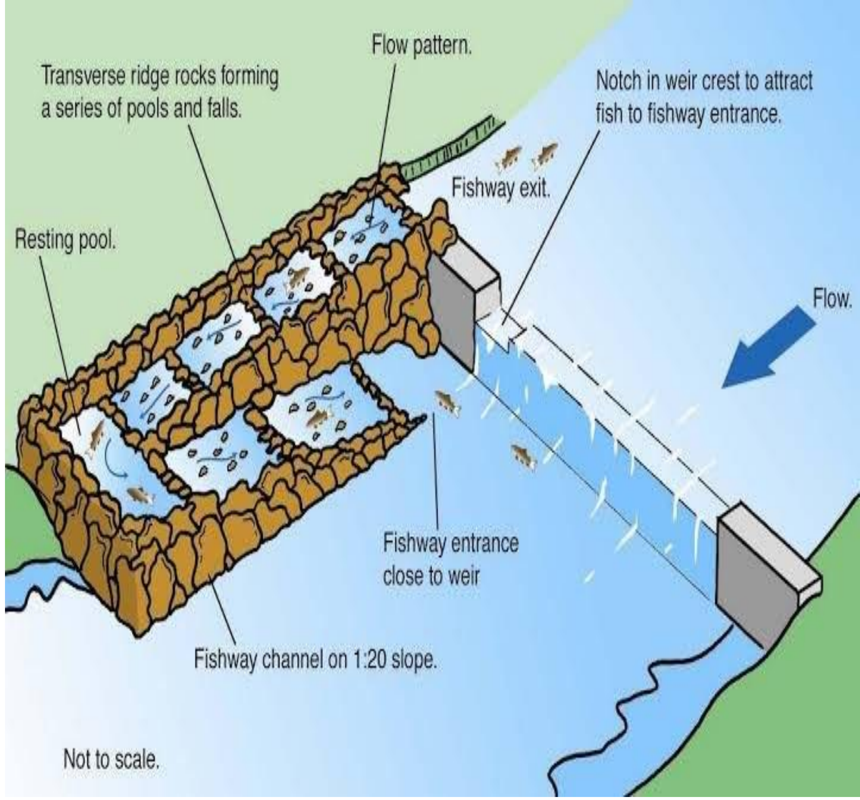
Accumulated toxic sediment damages benthic ecosystems and bioaccumulates through aquatic organisms, threatening downstream fisheries and human health.

Adverse Impacts: Fisheries & Biodiversity

1. Barrier to Fish Migration
The Padma Barrage may obstruct the natural migration routes of fish, especially migratory species.
Although fish passages are proposed, their effectiveness may vary and may not support all species equally.

2. Decline in Fish Population & Aquatic Biodiversity
Restricted migration can reduce breeding success and affect fish reproduction.
This may lead to a gradual decline in fish populations and overall aquatic biodiversity in the river system.

3. Impacts on Riverine & Floodplain Ecosystems
Changes in river flow and sediment transport may alter aquatic habitats and floodplain ecosystems.
Wetlands, spawning grounds, and other ecologically important habitats may be adversely affected.



Adverse Impacts: Waterlogging, Land Use & Resettlement

❖ Adverse Impact: Waterlogging

❑ Mechanism

The construction of the Padma Barrage may increase the water level in the upstream area, which may lead to the formation of a reservoir. Due to the formation of this reservoir, a backwater effect may occur, which may disrupt the natural drainage system of the surrounding low-lying areas. As a result, there may be a possibility of long-term waterlogging in the nearby areas.

❑ Impacts

- ➢ Agricultural land remains waterlogged, reducing crop production.
- ➢ Damage to houses, local roads, and other infrastructure.
- ➢ Disruption of local livelihoods.
- ➢ Negative impacts on the surrounding ecosystem.

❑ Mitigation Measures

- ➢ Construct adequate drainage channels and sluice gates.
- ➢ Restore natural canals and ensure integrated water management.
- ➢ Regular monitoring of waterlogging-prone areas.
- ➢ Proper maintenance of drainage systems.

Adverse Impacts: Waterlogging, Land Use & Resettlement

❖ Adverse Impact: Land Use Change

❑ Mechanism

The construction of the Padma Barrage may create a reservoir, causing agricultural land, char areas, and low-lying land to become permanently or seasonally submerged. As a result, existing land use may change, and agricultural land may be converted into reservoir or water bodies.

❑ Impacts

- Reduced agricultural production and food security.
- Loss of agriculture-based employment and farmers' income.
- Negative impacts on the local economy.
- Reduced social stability.

❑ Mitigation Measures

- Promote alternative livelihoods (e.g., floating agriculture and modern aquaculture).
- Provide technical training and support small enterprise development.
- Establish industries to create employment opportunities.
- Ensure fair compensation and planned land use (Land Use Planning).

Adverse Impacts: Waterlogging, Land Use & Resettlement

❖ Adverse Impact: Resettlement

❑ Mechanism

The construction of the barrage and reservoir may require land acquisition in designated areas. As a result, many families may lose their homes and properties and may need to relocate to other areas. This may lead to significant social, economic, and cultural changes within the affected communities.

❑ Impacts

- Loss of livelihoods and separation from familiar surroundings.
- Disruption of social networks and community ties.
- Limited access to education, healthcare, employment, and other basic services.
- Reduced overall quality of life.

❑ Mitigation Measures

- Implement an effective **Resettlement Action Plan (RAP)**.
- Ensure fair compensation at market value.
- Provide safe and adequate housing.
- Implement **Livelihood Restoration** programs for affected communities.

Adverse Impacts: Downstream, Regional & Transboundary Concerns

- ❑ Concise briefing on risks, regional dynamics, and international implications of the proposed second Padma Barrage — focused on downstream impacts, regional trends, and transboundary dispute pathways.

Agenda

- ❖ Downstream Impacts
- ❖ Regional & Political Context
- ❖ Dispute Resolution Options

❑ **Downstream Impacts —**

Large barrages can alter downstream flow magnitude and seasonality. Risks include reduced dry-season flows, altered flood pulses that sustain wetlands, and unpredictable release schedules that harm farmers and fisheries.

❑ Regional ,Water-sharing and Political Dimension

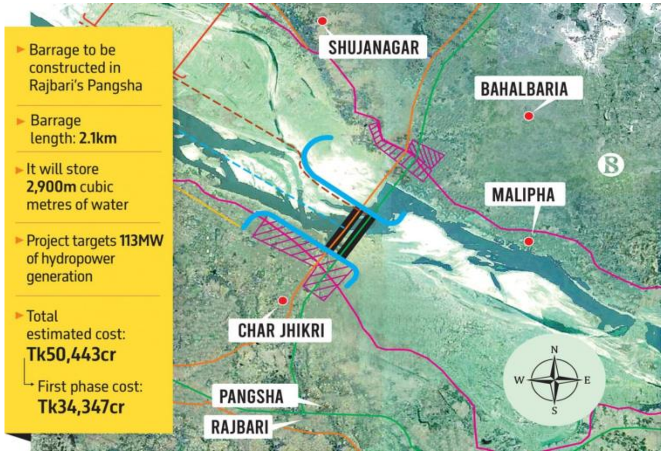
- Relationship to the Ganges Water Sharing Treaty & India's Farakka Barrage:

➢ The Padma River is part of the international Ganges River system shared by Bangladesh and India.

➢ The project's effectiveness depends on adequate dry-season inflow from upstream. It is closely linked to the 1996 Ganges Water Sharing Treaty and the operation of India's Farakka Barrage.

➢ Any upstream water management changes may influence the performance of the Padma Barrage.

➢ Strong regional cooperation is essential for sustainable water resource management.



▪ The Padma Barrage could create significant environmental and social risks if not supported by effective management, scientific monitoring, and regional cooperation.

❑ Paths for transboundary Dispute Resolution

- ❖ Strengthen bilateral dialogue through the Joint Rivers Commission (JRC).
- ❖ Ensure effective implementation of the Ganges Water Sharing Treaty (1996).
- ❖ Promote joint hydrological data sharing and scientific collaboration.
- ❖ If bilateral negotiations are unsuccessful, disputes may be addressed through international legal or diplomatic mechanisms, following international water law principles.
- ❖ Encourage regular stakeholder consultation and regional cooperation for long-term water security.

The Expert Divide: Supporters vs. Critics

As the Padma Barrage moved through approval, officials and engineers made the case for it — while geologists and former water officials raised early warnings.

Supporters

Government officials & BWDB engineers

- **Shahid Uddin Chowdhury Annie, Water Resources Adviser**
  "Essential for long-term water security and reducing vulnerability to seasonal flow fluctuations."
- **Malik Fida A Khan, Executive Director, CEGIS**
  Sees the barrage as necessary to meet the south-western region's dry-season water needs.
- **BWDB engineers**
  Say it will restore flow to the Gorai, Modhumati & Hisna rivers and expand the Ganges-Kobadak irrigation scheme.

VS

Critics

Geologists, academics & former officials

- **Md. Khalequzzaman, Prof. of Geology & Oceanography**
  Warns the project may "deepen Bangladesh's sediment, water and ecological crises" and calls it premature.
- **Mashfiqus Salehin, BUET Institute of Water & Flood Management**
  Warns it "may cause upstream erosion and downstream sedimentation, similar to Farakka."
- **Aktar Hossain, former Water Resources Secretary**
  Warns unchecked salinity intrusion could still affect parts of Khulna.

Sources: The Daily Star (thedailystar.net); BSS News (bssnews.net); TBS News (tbsnews.net); Dhaka Tribune (dhakatribune.com)

What the Critics Are Warning About

Beyond general caution, critics point to four specific, technically grounded risks — echoing patterns already seen at India's Farakka Barrage.

Sediment starvation of the delta

One of the world's most sediment-heavy rivers loses deposition downstream, risking accelerated coastal land loss — a pattern compared to the Mississippi Delta.

Upstream erosion, downstream sedimentation

BAPA warns riverbed rise could increase flooding and bank erosion along a 145 km stretch from Pangsha to Rajshahi.

Salinity creeping in from another direction

Diverting dry-season flow to the southwest may reduce flow in central rivers (e.g. Arial Khan), letting salinity move further inland via the Meghna estuary.

Risk of underperformance & lost leverage

If upstream flow from India stays inconsistent, the barrage may underdeliver — and could weaken Bangladesh's future negotiating position on Ganges water sharing.

Sources: The Daily Star (thedailystar.net); The Business Standard (tbsnews.net) — BAPA statement; bdnews24.com

Principles at Stake & the Scrutiny Debate

Beyond the technical debate, experts frame the controversy around two accountability principles — and a shared concern that approval outpaced public review.

Precautionary Principle

Environmental experts say ecological uncertainty — sediment loss, salinity shifts — justifies caution before locking in an irreversible structure.

Polluter-Pays Principle

Legal experts stress the need for clear accountability mechanisms so environmental damage from the project, if it occurs, has a responsible party.

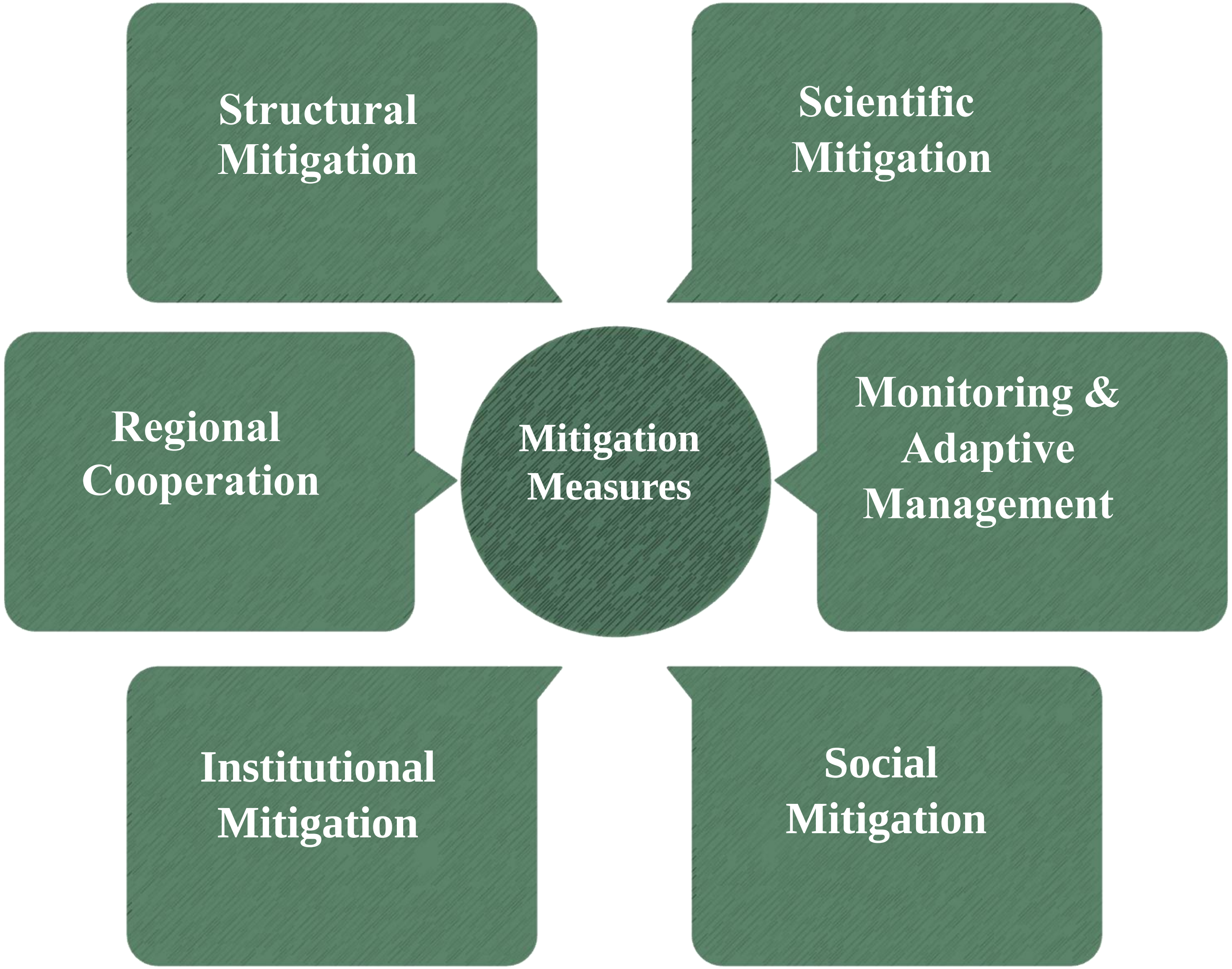
The rapid-approval critique

Environmental groups and civil-society voices say the project's approval moved forward with insufficient public consultation and technical review — and warn the project should be judged on ecological safety and downstream community rights, not engineering ambition alone.

> ***"Development cannot come at the cost of environmental collapse."*** — Sohanur Rahman, YouthNet Global

Sources: universalinstitutions.com; theclimatewatch.com; Vision IAS (visionias.in)

❑ Mitigation Measures & Impact Minimization Strategies



❑ Mitigation Measures & Impact Minimization Strategies

Structural Mitigation

- Properly engineered fish passages to preserve migration routes
- Scheduled sediment flushing and sluice operations
- Adaptive gate management responsive to seasonal flow

Scientific Mitigation

- Mandatory advanced hydrological modelling before construction
- Continuous sediment-transport modelling during operation
- Modelling outputs used to set gate-operation rules

Monitoring & Adaptive Management

- Continuous water quality, sediment and fisheries monitoring
- Pre-defined adaptive-management trigger thresholds
- Findings fed back into operating rules in real time

Social Mitigation

- Resettlement action plans with fair, timely compensation
- Livelihood-restoration programs for fishing and farming communities
- Precedent: Padma Bridge EIA (~231 ha of farmland affected)

Institutional Mitigation

- Independent environmental audits of construction and operation
- Transparent, ongoing public consultation with affected communities
- Phased implementation with built-in review checkpoints

Regional Cooperation

- Coordinated water-sharing dialogue with India on Ganges flows
- Addresses the upstream root cause, not only downstream symptoms
- Recourse to international forums if bilateral talks stall

Conclusion

The proposed Padma Barrage Project has the potential to improve water security, irrigation, navigation, hydropower generation, and salinity control.
However, it may also create significant environmental and social challenges, including sediment disruption, biodiversity loss, and community displacement.
Therefore, the project should be implemented with a balanced approach that ensures both development and environmental sustainability.

Lessons Learned:

Comprehensive and independent Environmental Impact Assessment (EIA) is essential before implementation.
Scientific decision-making should rely on advanced hydrological and sediment modelling.
Stakeholder participation and transparency are critical for long-term project success.
Continuous monitoring and adaptive management are necessary to minimize unforeseen impacts.

Recommendations

1. Strengthen the EIA review process through independent expert evaluation.

2. Implement continuous monitoring of water quality, sediment transport, and aquatic biodiversity.

3. Ensure fair compensation, proper resettlement, and livelihood restoration for affected communities.

4. Improve regional water cooperation and dialogue for sustainable transboundary river management.

5. Apply adaptive management strategies based on regular environmental monitoring results.

Closing Statement

"The Padma Barrage Project can become a milestone for Bangladesh only if economic development is achieved together with environmental protection and social responsibility."

Thank You